PROFESSIONAL ENGINEERING TRAINING DVISION
NATIONAL ENGINEERING CENTER
University of the Philippines

TRAINING ON
MODULE I: AGENTIC AI & LARGE LANGUAGE MODELS
Seminar Room A, Room 206-207, 2nd Floor, Juinio Hall, UP Diliman, QC

REMINDERS TO PARTICIPANTS

(1) TRAINING SCHEDULE

The training will be conducted on November 9, 10, 12, and 13, 2026, from 8:00am to 5:00pm. Topics to be covered each day are as follows:

SESSION DAYS	ACTIVITIES*
DAY 1 8:00 AM – 5:00 PM	**Opening Program (Reminders, Intro of Resource Person)** Lecture 1: Introduction to Agentic AI Project Case: AI Project Proposal Generation Presentation of Project Proposal **Lunch Break** Lecture 2: Deep Learning, Transformers, and LLM Foundations Project Case: Mapping the Business Process
DAY 2 8:00 AM – 5:00 PM	Lecture 3: Local LLMs, Prompts and Structured Outputs Case Study: LLM-Powered Information Extraction with Python and Project Case **Lunch Break** Lecture 4: Embeddings and Retrieval-Augmented Generation Case Study: Building A Local Organizational RAG System and Project Case
DAY 3 8:00 AM – 5:00 PM	Lecture 5: Orchestrating AI Agents: Tools, State, Memory, and MCP Case Study 5: Building a Single Tool-Using Agent with Python and Project Case **Lunch Break** Lecture 6: Designing Stateful Multi-Agent Systems Case Study: Designing a Multi-Agent System and Project Case
DAY 4 8:00 AM – 5:00 PM	Lecture 7: Agentic AI Assurance: Guardrails, Security, and Evaluation Case Study: Testing Agentic AI: Attacks, Guardrails, and Evaluation and Project Case **Lunch Break** Lecture 6: Large Language Model Operations (LLMOps) Project Case Work Final Project Presentation **Closing Program (Evaluation, Certificate-giving, Batch Picture-taking)**

*At the discretion of the Resource Person/Program Manager, topics/activities may change without prior notice.

(2) TRAINING MATERIALS

In support of a greener environment, UP NEC will only be providing soft copies of the learning materials. These will be sent through the email addresses you've indicated during registration. Upon receipt, kindly save the files in the laptops you will be using for the training.

Kindly take note that unauthorized copying, sharing, and distribution of the training materials are strictly prohibited, including, but not limited to, digital copies of handouts, assessments, and activities.

Email your required case study and project case work output, questions, and concerns to the assigned Training Manager.

(3) HOUSE RULES

During the training sessions, kindly observe the following:

a. Arrive before 8:00AM, so as not to miss out on important announcements and discussions.
b. Sign in the attendance sheet, once in the morning (upon arrival) and once in the afternoon (before the afternoon session starts).
c. Wear your IDs at all times (to be issued on the first day).
d. Keep mobile phones in silent mode to minimize distractions.
e. Actively participate in the discussions. If you have a question but you don't want to ask in front of others, ask it privately during break time. Please do not think any question you have is unimportant.
f. At the end of the training, you will be asked for your honest feedback to make this training better. Constructive criticism is appreciated and is the only way that we can improve our services.

(4) COURSE REQUIREMENTS

The training team will check the attendance based on those present during the sessions. To receive a Certificate of Completion, participants must attend at least 90% of the total training hours, in addition to the submission of all required outputs, and settlement of the training fee.

(5) NON-COMPLETION

Those who cannot complete the training program due to official or personal reasons should immediately inform the assigned Training Manager.

(6) OTHER REMINDERS

If you will be using a company-issued laptop, please ensure that you have installed the necessary software, followed the set-up guide, and that you can connect to the internet via Wi-Fi (password will be provided at the venue).

Snacks and lunch will be served at Lounge A (Room 208-209). Morning snack is at around 10am, lunch is at 12nn, and afternoon snack is at 3pm. You are encouraged to bring your own water containers. We have coordinated your food restrictions to the caterer; however, you may bring your own food in case the ones served do not meet your standards.

Parking space will be on a first-come-first-served basis. Smoking is strictly prohibited inside the UP NEC building.

Bring your own jackets in case the training room becomes too cold for comfort.

Always observe **CLAYGO (CLean As You GO)** after every training session day, and please make sure you do not leave any personal belongings behind.

UP NEC TRAINING
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UP NATIONAL ENGINEERING CENTER

Agentic Artificial Intelligence and Large Language Models

Data Science and Analytics Certification Program Module I



Modality & Venue: **Face-to-Face at UP NEC (Juinio Hall), UP Diliman, Quezon City**
Training Fee: **PHP 18,000.00**

This **module** equips participants with the foundational knowledge and practical skills to **design and develop Large Language Model (LLM) and Agentic Artificial Intelligence (AI) applications** using Python and open-source technologies. By the end of the course, participants will have developed and evaluated a working Agentic AI prototype aligned with real organizational use cases.

Build real-world AI applications using leading open-source technologies:

python | LangChain | LangGraph | Ollama | SBERT.net

Prerequisites:

1. Must have taken formal courses on **Modern Machine Learning** (UP NEC Module D: Predictive Analytics - Machine Learning Module or equivalent).
2. Must have taken **basic or intermediate Python programming** (UP NEC Module H: Python for Data Science and Analytics or equivalent).

Required Laptop Specifications:

1. At least an 8-core processor
2. 24-32 GB of RAM or unified memory
3. Minimum 80 GB of free SSD storage
4. **For Windows/Linux OS:** Optionally includes an NVIDIA GPU with at least 8 GB of VRAM
5. **For Mac**: Preferably uses an M1, M2, M3, M4, or newer Apple Silicon processor with at least 24 GB of unified memory.

Pre-registration Form
bit.ly/2026NECRegistration

Payment Guidelines
bit.ly/2026NECPaymentGuidelines

Connect, re-connect, and learn with us!

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